TRACE

- **Problem Statement ID - SIH26103**
- **Problem Statement Title - Use case on web-based integrated project-monitoring platform**
- **Theme - Smart Automation**
- **PS Category - Software**
- **Team ID -**
- **Team Name – Trace**

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AI-Powered Early-Warning System for Infrastructure Projects

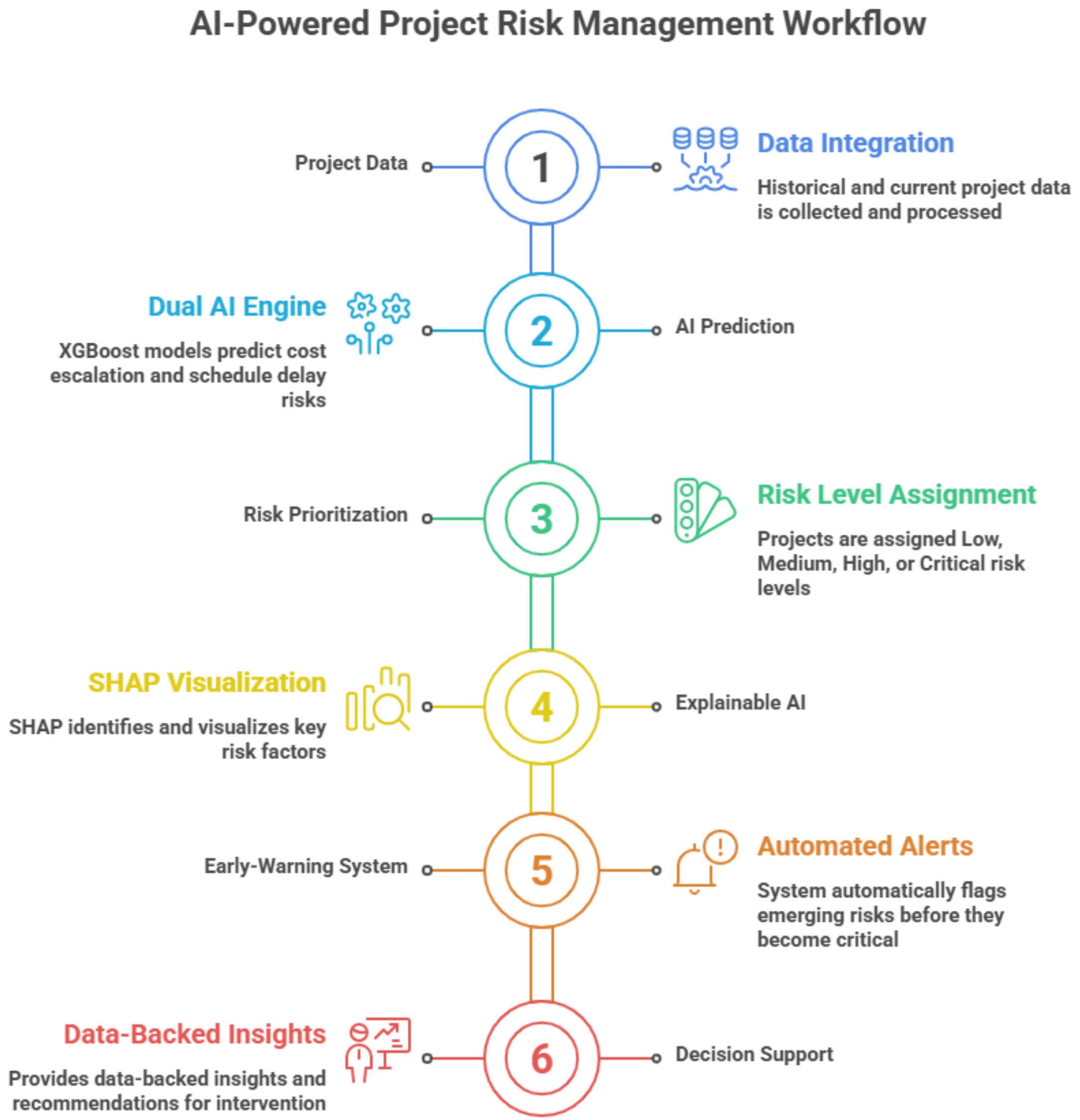
❖ Proposed Solution

How It Works:

Project Data → Data Processing → AI Prediction → Explainable Risk Score → Early Warning → Decision Support

- **Integrated Data Layer:** Processes historical and current project data such as cost, expenditure, physical progress, milestones and timelines.
- **Dual AI Prediction Engine:** Uses independent XGBoost models to predict cost escalation risk and schedule-delay risk.
- **Risk Prioritization:** Assigns every project a Low / Medium / High / Critical risk level, allowing authorities to focus on projects requiring immediate attention.
- **Explainable AI: SHAP** identifies and visualizes the key factors contributing to each prediction, making AI outputs transparent and interpretable.
- **Early-Warning System:** Automatically flags emerging cost, schedule and implementation risks before they become critical.
- **Decision Support:** Provides data-backed insights and recommended areas of intervention while keeping final administrative decisions with authorized officials.

Key Innovation:
From “What happened?” → “What is likely to happen?” → “Why?” → “Where should we intervene?”
Outcome: Faster risk identification • Better prioritization • Transparent AI • Proactive project governance



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TECHNICAL APPROACH

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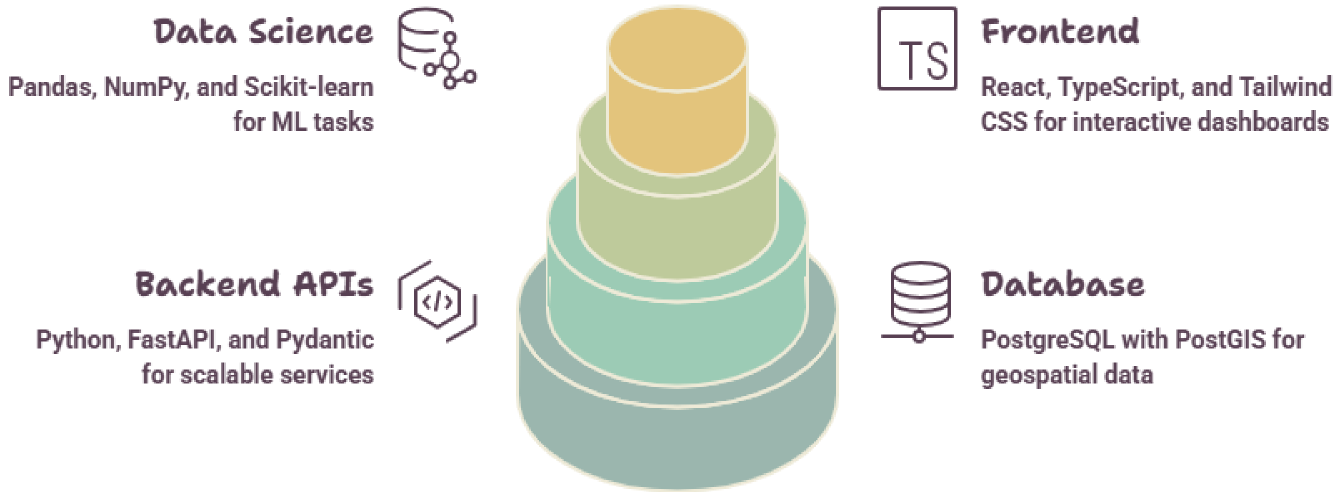
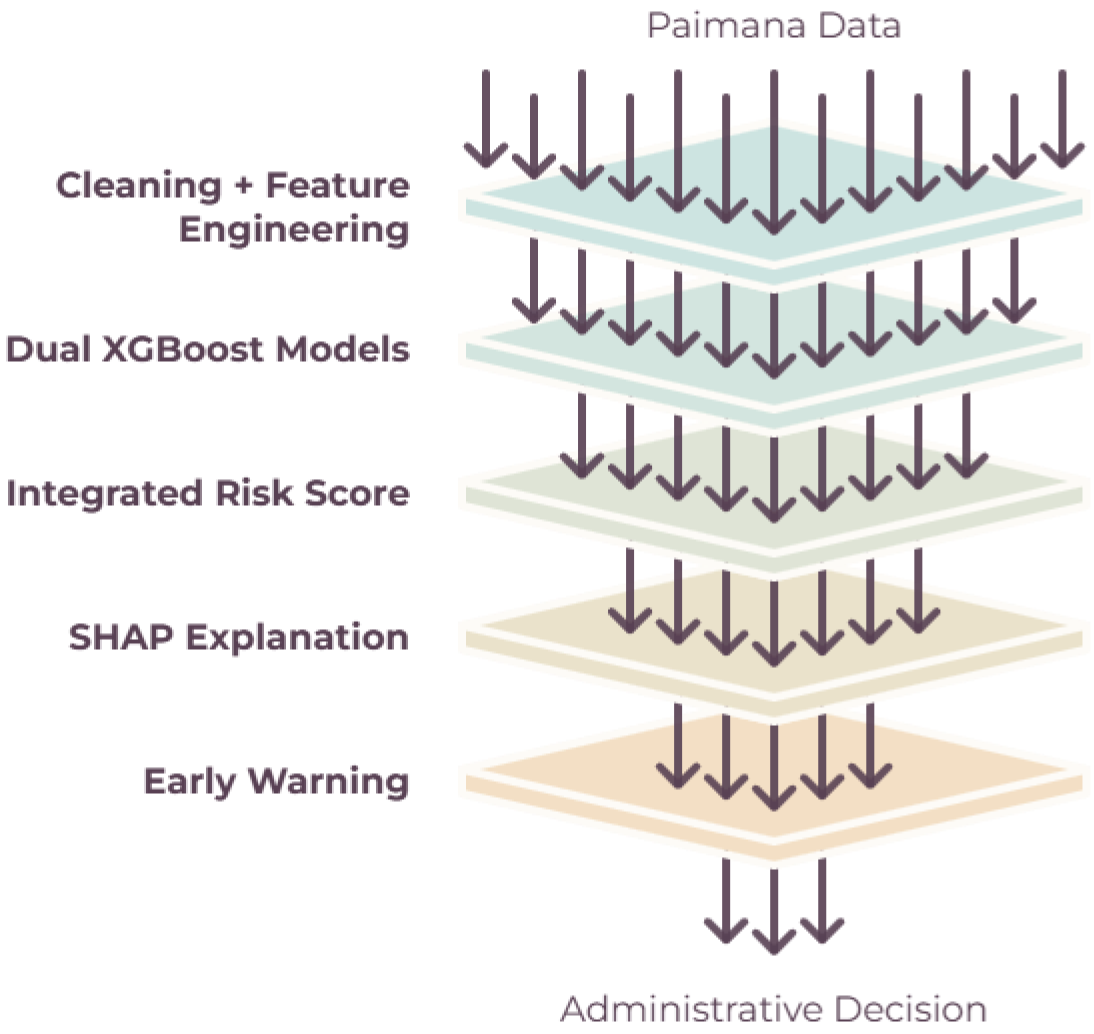
❖ Technical Approach:

- **Frontend:** React (Next.js 15), TypeScript, Tailwind CSS, Recharts, and Leaflet/MapLibre for interactive dashboards and geographic project visualization.
- **Backend / APIs:** Python, FastAPI, and Pydantic for scalable API services, integrated with **JWT Authentication**.
- **Database:** PostgreSQL hosted through **Supabase**, with **PostGIS** capabilities for storing and analyzing project geospatial data.
- **Data Science / ML:** Pandas, NumPy, Scikit-learn, XGBoost, and SHAP for data preprocessing, feature engineering, predictive modeling, risk classification, and explainable AI.





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FEASIBILITY AND VIABILITY

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Feasibility of the Idea

- Uses **open-source, industry-proven technologies**: Next.js, FastAPI, PostgreSQL, XGBoost and SHAP.
- Modular architecture enables independent development, testing and scaling.
- Historical **MoSPI / PAIMANA-style data** provides a foundation for model training and validation.
- Cloud-ready architecture supports deployment across departments and large project portfolios.

Potential Challenges & Risks

- **Data Quality:** Missing, inconsistent or outdated project information.
- **Data Availability:** Limited historical or labelled data for certain project categories.
- **System Integration:** Integration with existing government monitoring systems.
- **Security & Privacy:** Sensitive project and administrative data requires controlled access.

Mitigation Strategies

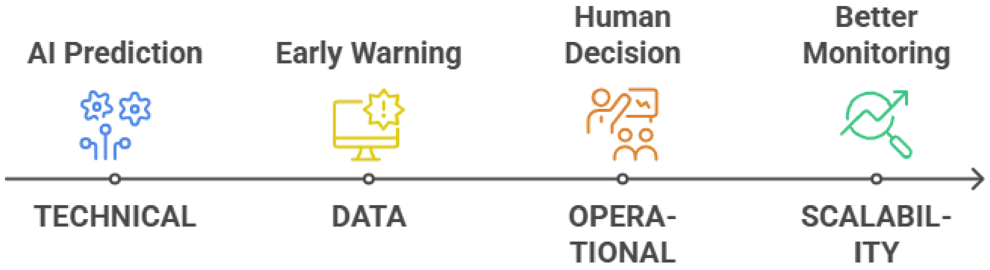
- Automated **data validation, cleaning and anomaly detection**.
- Time-aware feature engineering and continuous model evaluation.
- Compare **XGBoost with conventional statistical baselines** before deployment.
- Use **SHAP Explainable AI** to show the factors behind every prediction.
- Adopt **API-first, modular architecture** for integration with existing systems.
- Implement **JWT, audit logs and secure database access**.

Business Potential & Use Cases

- Primary Users
- MoSPI / Government Departments
- Project Monitoring Units
- Implementing Agencies
- Infrastructure Authorities
- Senior Decision-Makers

Key Use Cases

- Early detection of **cost & schedule overruns**
- Portfolio-level **project risk prioritization**
- Infrastructure project performance monitoring
- Sector-wise and geographic risk analysis
- Executive dashboards and automated reporting
- Natural-language project intelligence through LLM assistant



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IMPACT AND BENEFITS

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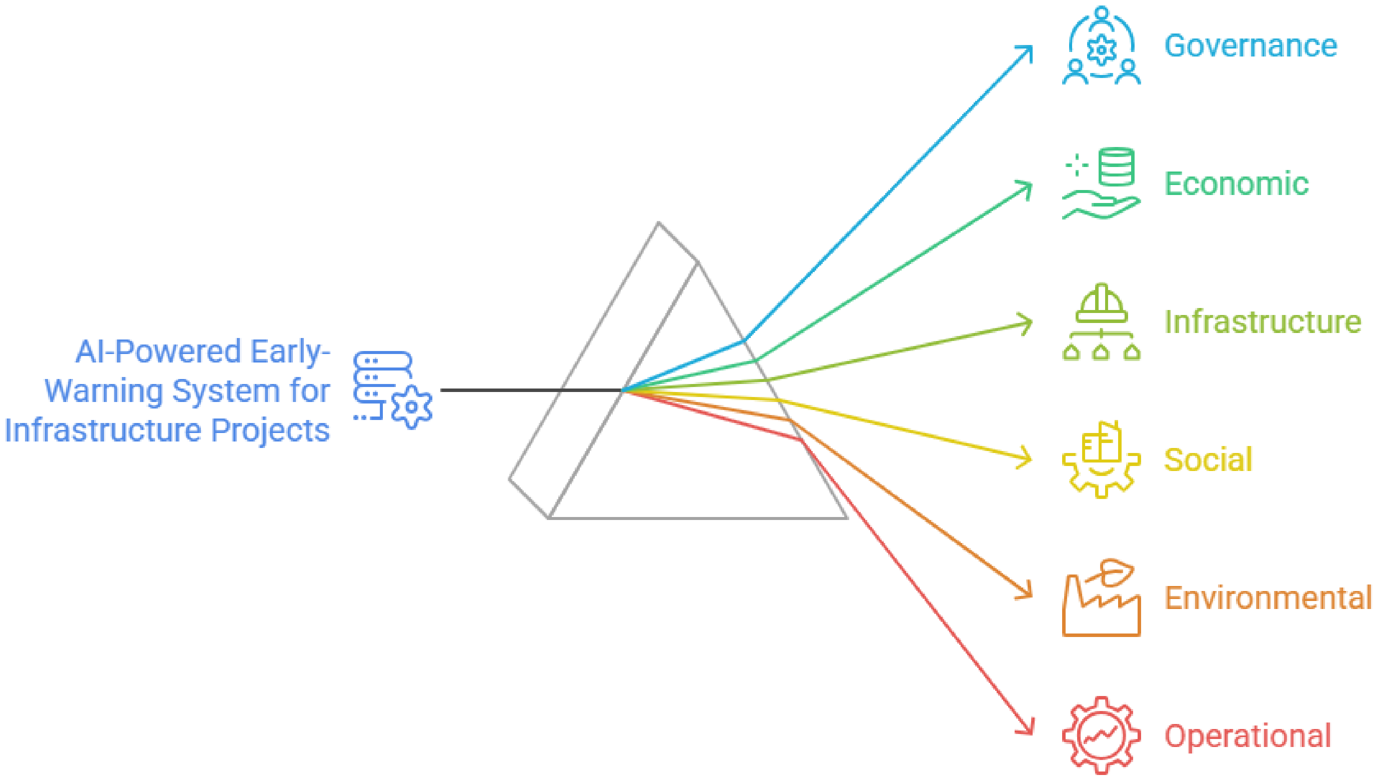

❖ Benefits of the Solution:

- **Social:** Supports timely delivery of public infrastructure and services
- **Economic:** Enables early detection of cost escalation and better resource prioritization
- **Operational:** Reduces monitoring effort and improves project oversight
- **Governance:** Promotes transparent, explainable and data-driven decision-making
- **Infrastructure:** Improves identification and management of project delays
- **Environmental:** Supports proactive monitoring of projects with environmental dependencies
- **Technological:** Converts existing project data into predictive and actionable intelligence

❖ Potential Impact on Target Audience:

- Enables **proactive monitoring** of infrastructure projects
- Helps officials identify **high-risk projects early**
- Reduces manual effort in project analysis and reporting
- Supports **faster, evidence-based administrative intervention**
- Provides transparent insights into **cost, schedule and implementation risks**

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RESEARCH AND REFERENCES

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Details / Links of Reference & Research Work

<u>Primary Data Source</u>

- MoSPI – PAIMANA Portal
- Monthly Flash Reports for Central Sector Infrastructure Projects
- PAIMANA Project Monitoring & Analytics Data

<u>Research / Algorithms</u>

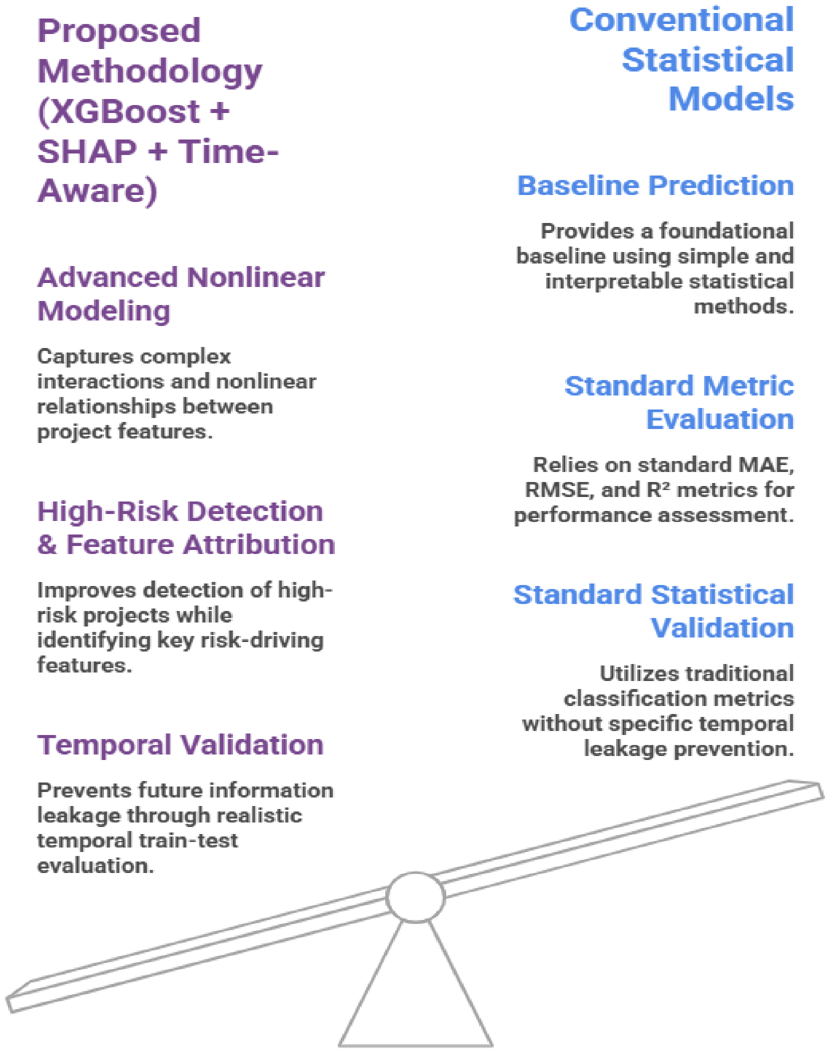
- XGBoost: Chen & Guestrin — *A Scalable Tree Boosting System*
- SHAP: Lundberg & Lee — *A Unified Approach to Interpreting Model Predictions*
- Scikit-learn: Model development, preprocessing & evaluation methods

<u>Reference Implementation</u>

Infrastructure-delay **XGBoost implementation** used strictly as a reference for model architecture and feature-engineering approaches.

<u>Key Research Considerations</u>

- Time-aware prediction
- Prevention of temporal data leakage
- Handling class imbalance
- Explainable risk prediction
- Model comparison & validation



Research & Model Evaluation Framework for SIH 2026

Highlight XGBoost + SHAP + Time-Aware Validation as the proposed methodology.